

Infrared Remote Control

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1. Introduction to Infrared Remote Control

Infrared remote control is a wireless, non-contact control technology with strong anti-interference capability, reliable information transmission, and low power consumption. However, it cannot transmit through obstacles. When the infrared remote control is pressed, it drives the infrared LED to blink at high speed (invisible to the naked eye). After the infrared receiver identifies it, the internal circuit processes through optical filtering, demodulation, and shaping to output a level signal, following the NEC protocol.

2. NEC Data Frame Format

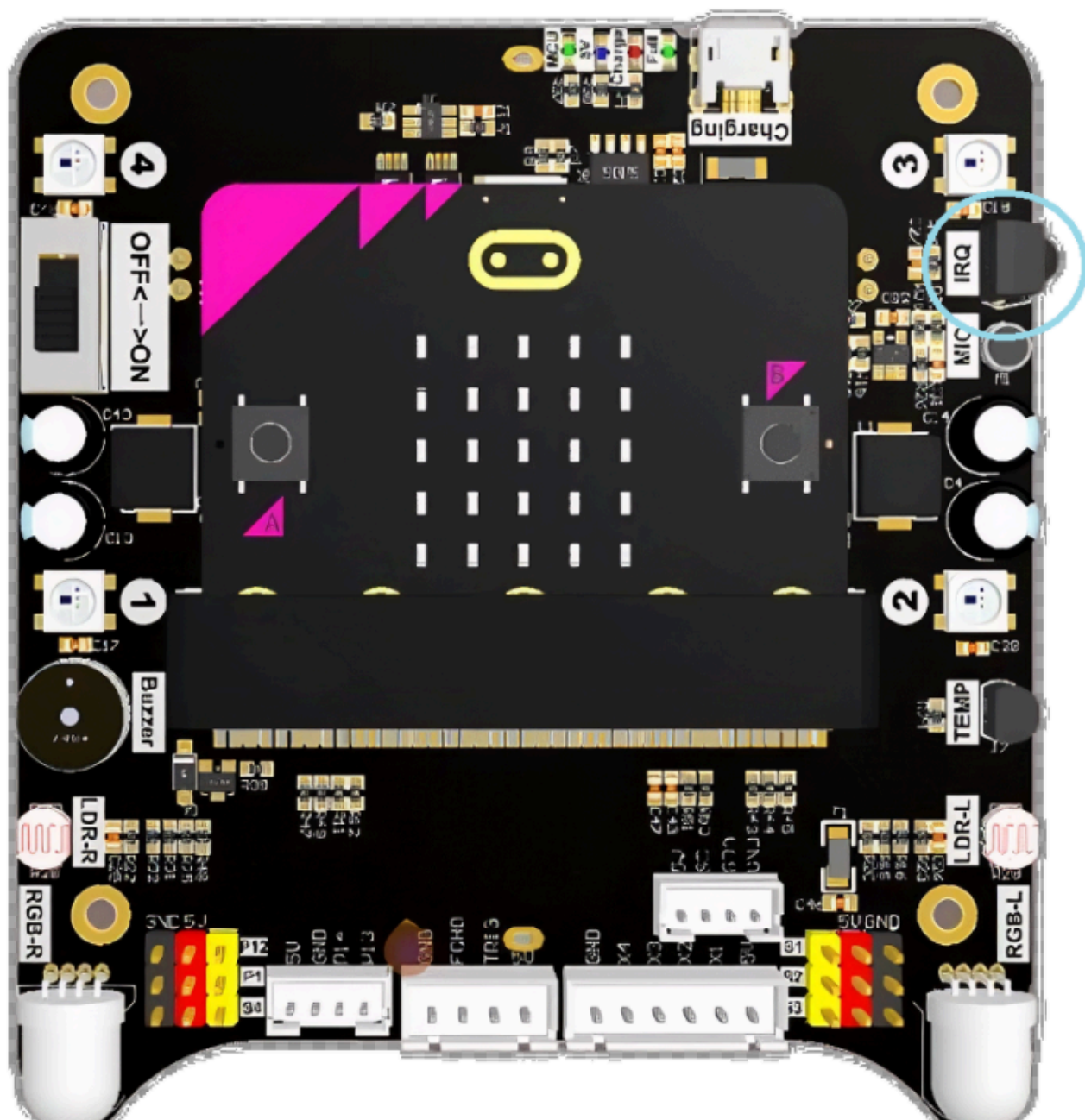
Each standard NEC protocol key press sends **4 bytes (32 bits)** of data, with the following structure:

Leader Code → Address Code → Address Inverse Code → Command Code → Command Inverse Code → Stop Bit

- **Address Code:** Identifies the device type (such as TV, DVD), preventing different devices from interfering with each other.
- **Address Inverse Code (~Address):** The bitwise inverse of the address code, used to verify the correctness of address transmission.
- **Command Code:** The specific key instruction (such as volume+, power off).
- **Command Inverse Code (~Command):** The bitwise inverse of the command code, used to verify the correctness of command transmission.

3. Precautions

1. You need to burn the infrared remote control program `2.Infrared_Remote.hex` to the Microbit.
2. There is a plastic film at the bottom of the infrared remote control from the factory. It needs to be removed before use.
3. When controlling, face the infrared receiver probe directly.



4. Infrared Remote Control Button Functions

User code: 00FF



Key Values	Actual Function	Key Values	Actual Function
 0x00	Stop motor and turn off lights	 0xB0	Control robotic arm move upward
 0x80	Control car forward	 0x08	Control robotic arm move left
 0x40	Control lights (Lighting effect changes with each press)	 0x88	Reset robotic arm
 0x20	Control car turn left	 0x48	Control robotic arm move right
 0xA0	Control buzzer	 0x28	Control robotic arm open gripper
 0x60	Control car turn right	 0xA8	Control robotic arm move downward
 0x10	Control car rotate left	 0x68	Control robotic arm close gripper
 0x90	Control car backward	 0x18	Grab block and lift robotic arm
 0x50	Control car rotate right	 0x98	Control robotic arm to gripping state
 0x30	Control car accelerate	 0x58	Put down block
 0x70	Control car decelerate	0xFF	Default Values